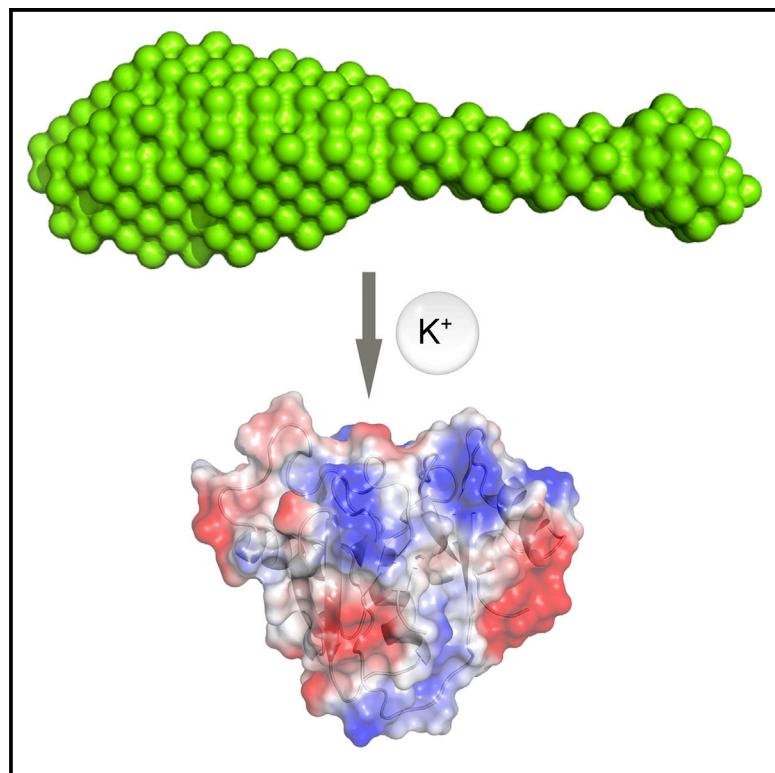


Structure

The Potassium Binding Protein Kbp Is a Cytoplasmic Potassium Sensor

Graphical Abstract



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In Brief

Ashraf et al. describe the discovery and structural analysis of a highly specific potassium binding protein, Kbp, from *E. coli*. Kbp is required for normal growth of *E. coli* at high K^+ concentrations. Structural analysis shows that K^+ binding induces a global conformational change in Kbp.

Highlights

- *E. coli* Kbp is a highly specific cytoplasmic potassium binding protein
- Kbp is required for normal growth of *E. coli* at high K^+ concentrations
- Both the BON and LysM domains of Kbp are required for full K^+ binding
- Structural analysis indicates global conformational changes on K^+ binding

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The Potassium Binding Protein Kbp Is a Cytoplasmic Potassium Sensor

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SUMMARY

Escherichia coli possesses a number of specific K⁺ influx and efflux systems that maintain an appropriate intracellular K⁺ concentration. Although regulatory mechanisms have been identified for a number of these transport systems, the exact mechanism through which K⁺ concentration is sensed in the cell remains unknown. In this work we show that Kbp (K⁺ binding protein, formerly YgaU), a soluble 16-kDa cytoplasmic protein from *Escherichia coli*, is a highly specific K⁺ binding protein and is required for normal growth in the presence of high levels of external K⁺. Kbp binds a single potassium ion with high specificity over Na⁺ and other metal ions found in biological systems, although, in common with K⁺ transporters, it also binds Rb⁺ and Cs⁺. Dissection of the K⁺ binding determinants of Kbp suggests a mechanism through which Kbp is able to sense changes in K⁺ concentration over the relevant range of intracellular K⁺ concentrations.

INTRODUCTION

Potassium is the major intracellular cation and is concentrated in the cytoplasm by transport systems that have a high specificity for K⁺ over other monovalent cations such as Na⁺. Due to the central role of K⁺ in key homeostatic mechanisms such as maintenance of intracellular pH and osmolarity, the cytoplasmic K⁺ concentration is closely controlled (Booth, 1985; Weiden et al., 1967). However, despite the identification and detailed characterization of K⁺ transporters and channels in *Escherichia coli*, our understanding of how intracellular levels of K⁺ are sensed and how an appropriate K⁺ concentration is maintained is incomplete. For bacteria such as *E. coli*, which are able to colonize a range of environments that vary widely in pH, osmotic potential regulating the intracellular K⁺ concentration is likely to provide a significant challenge.

In the cytoplasm of *E. coli* during normal growth the K⁺ concentration is approximately 250 mM, although this may

vary significantly depending on exact environmental conditions (Weiden et al., 1967). Potassium influx forms a primary response upon osmotic upshock as bacterial cells begin to lose turgor pressure due to the efflux of water. The accumulation of potassium glutamate inside the cytosol leads to several adaptive changes resulting in the reprogramming of cellular gene-expression profiles, transport, and de novo synthesis of neutral osmoprotectants. Changes to cellular transcription have been shown to be mediated directly by potassium glutamate, which is able to directly inhibit RNA polymerase (Gralla and Vargas, 2006). In addition, potassium glutamate has been shown to directly stimulate transcription from an RpoS-dependent promoter (Lee and Gralla, 2004). The RpoS regulon in *E. coli* consists of a number of stress response genes which are upregulated by osmotic stress (Vijayakumar et al., 2004).

E. coli contains a number of K⁺ transport systems, with active uptake mediated by the Trk, Ktr, and Kup transporters and active efflux by the Kef transporter (Bossemeyer et al., 1989a, 1989b; Nakamura et al., 1998; Roosild et al., 2010). In addition, Kch, an *E. coli* homolog of eukaryotic K⁺ channels, has been shown to form an active potassium-conducting channel in vivo, although the physiological conditions under which the activity of K⁺ channel is required have not been established (Kuo et al., 2003). K⁺ transport systems are able to readily discriminate between K⁺ and Na⁺, which differ in radius by only 0.38 Å, and the basis for the specificity of K⁺ channels has been widely investigated, both experimentally and theoretically (Doyle et al., 1998; Noskov and Roux, 2006; Yu et al., 2010; Zhou et al., 2001).

The cytoplasmic protein Kbp (K⁺ binding protein, formerly YgaU) from *E. coli* is one of a small number of proteins that are strongly induced in response to osmotic stress, due to the addition of sodium chloride, and is a member of the RpoS regulon (Vijayakumar et al., 2004). Kbp is a 16-kDa protein that is predicted to be located in the cytoplasm, and contains an N-terminal BON (bacterial OsmY and nodulation) domain and a C-terminal LysM (lysine motif) domain. The LysM domain is frequently associated with peptidoglycan binding, and changes in the cellular level of Kbp (YgaU) have been shown to modulate peptidoglycan cross-linking (Bateman and Bycroft, 2000; Bernal-Cabas et al., 2015). The BON domain is predicted to be a membrane-associated phospholipid binding domain, although the experimental evidence to support this assertion is controversial (Teriete et al., 2010; Yang et al., 2011; Yeats and Bateman, 2003). However,

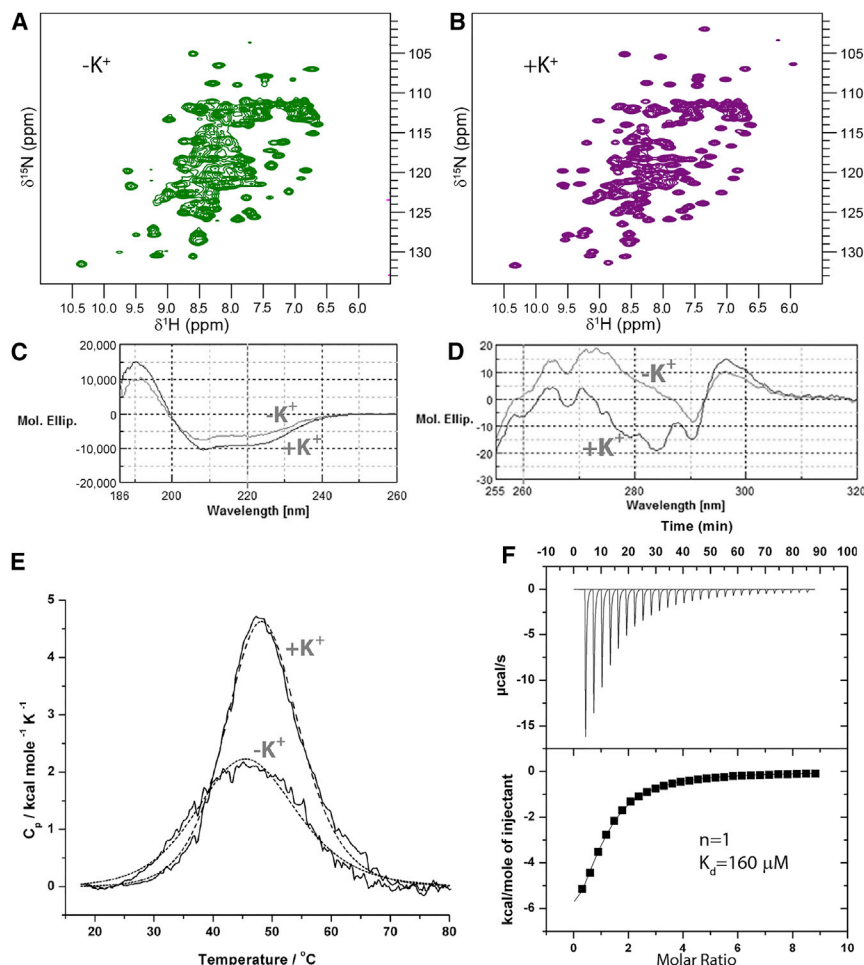


Figure 1. K^+ Binding Leads to Stabilization and Structural Changes in Kbp

(A and B) ^{15}N -HSQC NMR spectra of 200 μM Kbp in 20 mM NaPi (pH 7.2) in (A) the absence and (B) the presence of 5 mM KCl.

(C and D) Near-UV (C) and far-UV (D) CD shows changes to secondary and tertiary structure of Kbp on K^+ binding. Near-UV CD was performed in 50 mM Tris and 50 mM NaCl (pH 7.5) $\pm$ 10 mM KCl at a protein concentration of 100 μM in a quartz cuvette of path length 0.5 cm. Far-UV CD was performed at a protein concentration of 30 μM in 20 mM Tris and 20 mM Na_2HPO_4 (pH 7.5) $\pm$ 5 mM K_2HPO_4 in a quartz cuvette of path length 0.02 cm. (E) DSC shows that Kbp is stabilized on binding K^+ . Solid lines show experimental thermograms for Kbp in the presence and absence of K^+ . Dashed lines show the fit with a two-state unfolding model. Binding of K^+ gives rise to an increase in the T_m from 45.9°C to 48.4°C and an increase in the enthalpy of unfolding from 50.2 to 77.8 kcal mol $^{-1}$. Experiments were performed in 50 mM Tris and 50 mM NaCl (pH 7.5) $\pm$ 1.27 mM KCl at a protein concentration of 64 μM .

(F) K^+ binding to Kbp monitored by ITC. The upper panel shows sequential heat pulses, after subtraction of dilution heats, on titration of KCl into Kbp in 50 mM Tris and 50 mM NaCl (pH 7.5) at 25°C. The lower panel shows integrated heat data fit with a one-site binding model. These data show that Kbp binds a single K^+ ion, the equilibrium dissociation constant for the Kbp- K^+ complex being 161 μM (Table S1).

See also Figures S1, S2, and S4.

there is direct experimental evidence that Kbp is peripherally associated with the cytoplasmic face of the inner membrane (Papanastasiou et al., 2013). BON domains are predicted for two other *E. coli* proteins; OsmY and YraP. Although the exact roles of these proteins are not fully understood, expression levels of OsmY are upregulated in response to osmotic stress, and YraP is an outer membrane lipoprotein that is likely involved in outer membrane biogenesis (Onufryk et al., 2005; Yim and Villar-ejo, 1992). In this work we show that the cytoplasmic protein Kbp is required for normal growth in the presence of K^+ concentrations in excess of 1 mM and under conditions of salt-induced osmotic stress. Biophysical and structural analysis shows that Kbp binds a single potassium ion and that K^+ binding induces a large conformational change in the protein. The isolated BON domain is able to bind potassium with an affinity of approximately 20 mM, suggesting a mechanism by which Kbp may act as a sensor of cytoplasmic K^+ concentration.

RESULTS

Kbp Is a K^+ Binding Protein

In *E. coli*, elevated levels of Kbp have been reported in response to osmotic stress associated with high levels of sodium chloride (0.4 M), but not organic osmolytes (Weber et al., 2006). In addition,

kbp (*ygaU*) has been shown to be regulated by the alternative sigma factor RpoS that regulates the expression of many stress response genes (Vijayakumar et al., 2004). These results suggest that Kbp may serve a key role in adaptation and stress responses.

Our initial experiments showed that Kbp could be readily over-expressed in *E. coli* BL21(DE3) and purified by nickel-affinity chromatography and gel filtration. Gel filtration indicated that this protein was present as a single species that eluted at a position consistent with the protein being monomeric. However, ^{15}N -heteronuclear single-quantum coherence (HSQC) spectra of ^{15}N -labeled Kbp purified by gel filtration and recorded in 20 mM sodium phosphate (pH 7.2) were poorly resolved, with fewer than the expected number of crosspeaks, and with broad linewidths and a lack of chemical-shift dispersion for many of the peaks that remained, indicating that a significant proportion of Kbp lacks a well-defined 3D structure under these conditions (Figure 1A). Serendipitously, we discovered that spectra recorded in buffer containing potassium ions were well dispersed and well resolved with crosspeaks of uniform line shape, indicating that K^+ ions are required to stabilize Kbp structure (Figure 1B). Thus, spectra recorded in sodium phosphate buffer with 5 mM KCl were well dispersed, but those recorded with even 250 mM Na^+ were not. In addition, differential scanning calorimetry (DSC) showed a

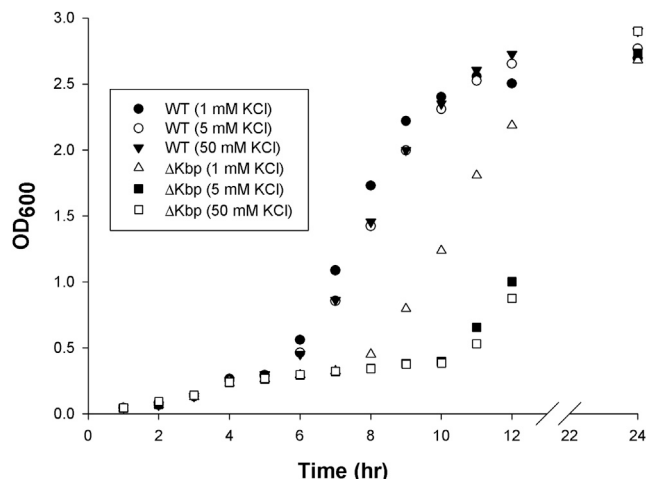


Figure 2. *E. coli* Lacking Kbp Shows Retarded Growth at High Concentrations of K^+

Wild-type and Δkbp *E. coli* BW25113 cells were grown at 37°C in defined sodium phosphate-based media with added KCl at 1, 5, and 50 mM. At higher K^+ concentrations (1, 5, and 50 mM) there was little variation in growth rate with K^+ concentration for the wild-type strain; however, the Δkbp strain showed a delay of around 2 hr in entering exponential phase at 1 mM KCl and around 5 hr at 5 and 50 mM KCl. See also Figure S3.

marked increase in the thermodynamic stability of Kbp in the presence of K^+ , and far- and near-UV circular dichroism (CD) spectra of Kbp recorded in the presence and absence of K^+ also indicated that Kbp undergoes significant structural changes on K^+ binding (Figures 1–1E).

To further investigate the binding of K^+ and other monovalent cations by Kbp, we performed isothermal titration calorimetry (ITC) experiments. Titration of KCl into a solution of Kbp in 50 mM Tris and 50 mM NaCl (pH 7.5) at 298 K showed that formation of the Kbp- K^+ complex is enthalpy driven, $\Delta H = -9.8 \text{ kcal mol}^{-1}$, and entropically disfavored, $\Delta S = -4.6 \text{ kcal mol}^{-1}$ (Figure 1F). This is consistent with ordering of the Kbp structure on K^+ binding inferred from our nuclear magnetic resonance (NMR) and DSC experiments. The K_D for the Kbp- K^+ complex is approximately 160 μM , with a single K^+ binding site per Kbp molecule (Table S1). No detectable heats of binding were observed on titration of sodium chloride into Kbp in 50 mM Tris and 50 mM LiCl (pH 7.5). Taken together with the data presented in Figure 1, this suggests that Kbp has little affinity for Na^+ . Thus, Kbp is able to effectively discriminate between K^+ and the chemically similar, but smaller alkali metal ion, Na^+ . ITC experiments with the larger alkali metal ions Rb^+ and Cs^+ showed that Kbp forms a complex with both these ions, with the affinity of Kbp for Rb^+ similar to that of K^+ , and the affinity for Cs^+ more than 10-fold lower (Figure S1 and Table S1). ITC studies with the K^+ channel, KcsA, from *Streptomyces lividans*, have shown that this protein displays an affinity for K^+ similar to that of Kbp. In addition, KcsA was shown to bind Rb^+ and Cs^+ but not Na^+ (Lockless et al., 2007). The soluble protein Kbp therefore displays cation specificity similar to that of a K^+ channel. Outside the alkali metal ions, Kbp binds NH_4^+ , which is also a substrate of potassium channels, with an affinity close to that for Cs^+ (Figure S1 and Table S1). However, ITC was unable to detect any

heat associated with binding on titration of the primary amine-containing compounds spermidine and putrescine.

In addition, we monitored K^+ association with Kbp under pseudo-first-order conditions using stopped-flow fluorimetry to follow the rate of Kbp- K^+ complex formation. Mixing Kbp with excess KCl resulted in the quenching of intrinsic Trp fluorescence (Figure S2A). The data were fitted with a single-exponential model, with k_{obs} increasing linearly with increasing KCl concentration, indicating that k_{obs} describes the bimolecular association step (Figure S2B). Values for the association rate constant (k_{on}) and dissociation rate constant (k_{off}) of $6.27 \pm 0.15 \times 10^4 \text{ M}^{-1} \text{ s}^{-1}$ and 7.89 s^{-1} were determined, respectively, from a plot of k_{obs} versus $[KCl]$ (Figure S2B). Calculation of K_D for the Kbp- K^+ complex from the ratio k_{off}/k_{on} gave a value of 126 μM , which agrees closely with the value determined by ITC under the same conditions. The dissociation rate constant showed little variation with increasing ionic strength to 0.3 M NaCl (8.97 s^{-1}) and a 3-fold increase with 0.6 M NaCl (17.3 s^{-1}). However, we saw a large decrease in the association rate constant with increasing ionic strength: k_{on} was reduced to $2.0 \times 10^4 \text{ M}^{-1} \text{ s}^{-1}$ in 0.3 M NaCl and $5.7 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$ in 0.6 M NaCl (Figure S2bc). Consequently, the K_D of the Kbp- K^+ complex increases approximately 30-fold on increasing the salt concentration from 50 mM to 600 mM NaCl, with K_D values of 0.1, 0.4, and 3 mM determined by stopped-flow fluorimetry at 50, 300, and 600 mM NaCl, respectively. Thus, at high ionic strength, which may better represent physiological conditions, the K_D for the Kbp- K^+ complex is considerably higher than at low ionic strength.

Kbp Is Required for Normal Growth at a Range of Elevated K^+ Concentrations

Since we have established that Kbp is a K^+ binding protein and its production has previously been shown to be upregulated on osmotic stress with NaCl, we hypothesized that the soluble cytoplasmic protein Kbp could be directly involved in K^+ homeostasis. Despite intensive research into K^+ transport, no protein has been identified that is a direct sensor of cytoplasmic K^+ concentration. To determine whether Kbp is involved in K^+ homeostasis, we compared the ability of the wild-type and Δkbp strains to grow at a variety of K^+ concentrations in a defined sodium phosphate-based minimal medium with KCl added at a range of concentrations. At low K^+ concentrations (0.05 and 0.01 mM KCl) there was no difference in the growth rates of the wild-type and Δkbp strains (Figure S3). At higher K^+ concentrations (1, 5, and 50 mM) the wild-type strain showed little variation in growth rate with K^+ concentration (Figure 2). However, the Δkbp strain showed a delay of around 2 hr in entering exponential phase at 1 mM KCl and a delay of around 5 hr at 5 and 50 mM KCl. Thus the loss of Kbp delays growth with increasing K^+ concentration (1–5 mM). Taken together, the specific binding of K^+ by Kbp and the K^+ -sensitive growth phenotype of cells lacking Kbp suggests that this cytoplasmic protein likely plays a key role in K^+ homeostasis, perhaps sensing cytoplasmic K^+ concentration. However, the K_D for the Kbp- K^+ complex seems incompatible with a role in cytoplasmic K^+ sensing, where the concentration of K^+ is typically around 200 mM. One possibility is that the affinity of Kbp for K^+ is modulated by protein-protein interactions or a competing non-protein ligand.

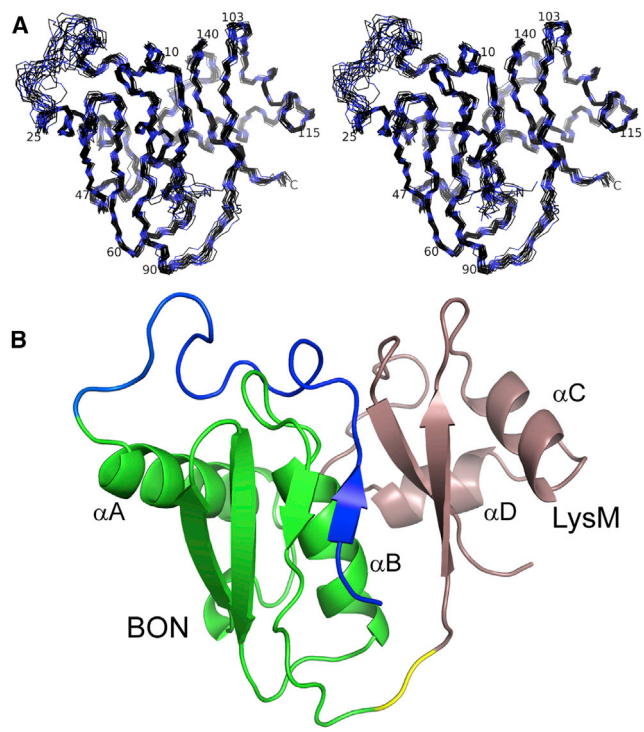


Figure 3. The Structure of K⁺-Saturated Kbp
(A) The ensemble of Kbp structures determined by NMR spectroscopy shown superimposed in backbone representation. The termini and a selection of residues are labeled and numbered.
(B) Cartoon representation of the Kbp structure with the BON and LysM domains colored green and brown, respectively. The N-terminal extension is shown in blue and the interdomain linker in yellow.

Full K⁺ Binding Requires Both the BON and LysM Domains of Kbp

To begin to dissect the K⁺ binding determinants of Kbp, we purified the isolated BON and LysM domains of Kbp. Near-UV CD indicated that the isolated BON domain undergoes a conformational change in the presence of K⁺ whereas the isolated LysM domain does not (Figure S4). In addition, ITC showed enthalpically driven complex formation between the BON domain and K⁺, whereas no heat was liberated on titration of KCl with the isolated LysM domain (Figure S4). However, ITC also indicated that K⁺ forms a substantially weaker complex with the isolated BON domain than with Kbp. The estimated K_D for the BON-K⁺ complex is ≥20 mM, in excess of 100-fold greater than for the Kbp-K⁺ complex (160 μM), suggesting that although the major interaction site lies within the BON domain, further interactions that stabilize the complex of K⁺ with Kbp involve its LysM domain. To monitor the conformational changes undergone by the BON domain on K⁺ binding, we produced ¹⁵N-labeled BON domain and monitored the changes to its ¹⁵N-HSQC spectra upon the addition of K⁺. In the absence of K⁺, ¹⁵N-HSQC spectra of the BON domain were poorly resolved, indicating that the isolated domain is largely unfolded. However, on addition of K⁺ the ¹⁵N-HSQC spectra improved, and at 270 mM K⁺ were well dispersed, indicative of a folded protein (Figure S4). In intact Kbp, residues in both the BON and LysM domains shift on K⁺ addition. Taken together, these data indicate

Table 1. Parameters for the Structure of Kbp Determined by NMR Spectroscopy

NOE Distance Restraints	Value
NOE restraints	5,626
Ambiguous	2,741
Unambiguous, of which	2,885
Intra-residue	1,237
Inter-residue	
Sequential (i – j = 1)	715
Medium-range (1 < i – j < 5)	371
Long-range (i – j ≥ 5)	562
Violations per structure >0.5 Å	0.05 ± 0.218
Violations per structure >0.3 Å	2.8 ± 1.66
Distance restraint rmsd (Å)	0.023 ± 0.001
Other Restraints	
Dihedral angle restraints	162
Parameter rmsd from idealized geometry	
Bonds (Å)	3.9 × 10 ^{−3} ± 7.90 × 10 ^{−5}
Angles (°)	0.49 ± 0.0097
Impropers (°)	1.39 ± 0.057
Coordinate RMSD ^{a,b} (Å)	
Backbone heavy atoms	0.699 ± 0.151
All heavy atoms	1.047 ± 0.145
Ramachandran statistics ^b (%)	
Most favored	84.3
Additionally allowed	13.0
Generously allowed	0.9
Disallowed	1.8

Average statistics were calculated from the 20 water-refined structures in the Kbp ensemble. The number of violations is shown as the average and SD per structure. Rmsd, root-mean-square deviation.
^aTo the unbiased mean structure.
^bResidues 2–149.

that the major K⁺ binding site is located in the BON domain and that this initial complex is stabilized by interaction with the LysM domain.

Solution Structure of Kbp-K⁺

To attempt to gain further insight into K⁺ binding by Kbp, we determined the structure of K⁺-saturated Kbp in solution by NMR spectroscopy (Figure 3 and Table 1). Kbp-K⁺ forms a compact, globular, heart-shaped structure (approximate dimensions 50 Å wide by 40 Å high by 30 Å deep) with the BON and LysM domains packed against one another. The N-terminal BON domain (residues 23–91) adopts an α-β-α-β topology forming a three-stranded mixed β sheet with α helix αB linking the second and third parallel strands, as seen in other BON domains (Teriete et al., 2010; Yang et al., 2011; PDB: 4qo6). However, the third strand is perturbed at N86 to accommodate a hydrogen bond between the amide of V87 and the side chain of D85. Helix αA (residues 24–38) packs across the C-terminal ends of the first strand and is joined to the first strand by a linker that incorporates another helical turn. Residues 1–23 augment the BON domain structure, with residues 2–6 forming a fourth,

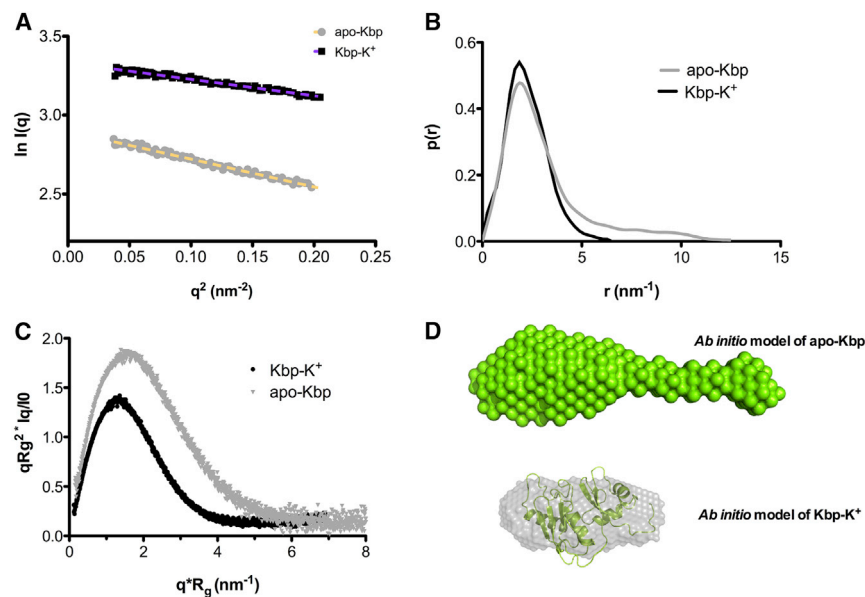


Figure 4. Conformational Changes in Kbp on K^+ Association Determined by SAXS

(A) Guinier analysis of apo- and K^+ -bound form shows a decrease in the radius of gyration (R_g) in the presence of K^+ as indicated by the reduced slope of the scattering intensity in the Guinier region of the SAXS curve. Extrapolation of the scattering intensity to zero concentration does not show any particle aggregation (data not shown).

(B) The distance distribution function for Kbp indicates a significant decrease in the maximum particle dimension (D_{max}) and a transition from an extended (12.5 nm) to a compact (6.2 nm) molecule in the presence of K^+ .

(C) The normalized Kratky plot for Kbp in the presence of K^+ (black) reveals a single peak with no obvious indications of flexibility, whereas the apo form of the protein (gray) exhibits a broader peak consistent with an increase in disordered content and an overall asymmetrical conformation, in accordance with the distance distribution function.

(D) The ab initio model of Kbp in the presence of K^+ overlaid with its NMR structure shows excellent agreement between the two structures. The apo form of the protein exhibits an elongated shape in solution.

See also Figure S5 and Table S2.

short β strand antiparallel to residues 81–85, and residues 7–23 forming a stretch of irregular structure that encloses the loop formed by residues 76–80. Residues 96–149 form a structure similar to those of previously determined LysM domains (Bate-man and Bycroft, 2000; Bielnicki et al., 2006; Koharudin et al., 2011; Sanchez-Vallet et al., 2013), with a two-stranded antiparallel β sheet enclosing one side of a small hydrophobic core whose other side is bounded by two, almost perpendicular α helices. However, the linker from the second helix of the LysM domain to the C-terminal β strand in Kbp lacks any of the helical character seen in this part of other LysM domains.

The two domains are tethered to one another by a short, ordered linker (residues 90–95) that caps the N terminus of αB , with a hydrogen bond forming between the amide group of Gln64 and the carbonyl of Thr91. The LysM domain packs principally against the opposite side of αB from the BON domain's β sheet, with the LysM β sheet facing the BON domain forming additional contacts to the N-terminal strand, the C-terminal end of αA , and its adjacent loop. The interface between the domains is maintained by a hydrophobic core involving F4, F6, T38, I40, L71, V72, and A73 from the BON and F98, T100, P130, M131, and V143 from the LysM domains, respectively. In addition there is a buried salt bridge between E65 and R145 close to the interdomain linker, while at the other end of the interface, N76 forms two hydrogen bonds to the backbone of V143 and Q142 is sandwiched between N76 and Y139. Q142, N76, and Y139 line the sides of a small pocket between the domains that is also bounded by E12, H34, M131, L132, and K133.

Comparison of Kbp- K^+ and apo-Kbp

To determine the extent of conformational changes in the context of the whole protein on K^+ binding, we used small-angle X-ray scattering (SAXS) and sedimentation velocity analytical ultracentrifugation data to model apo-Kbp and K^+ -bound Kbp.

High-quality SAXS data were acquired for a range of concentrations of Kbp in the presence and absence of K^+ (Figure 4A). Guinier analysis of these data revealed a significant change in the radius of gyration (R_g): 2.28 nm for the apoprotein and 1.79 nm for the K^+ -bound form. The reduction in R_g suggests a marked compaction of the overall Kbp structure upon K^+ binding. The $p(r)$ function describes the paired set of distances between all the electrons within the protein and allows detection of any conformational changes (here, in response to K^+). The maximum particle diameter (D_{max}) of Kbp in the presence of K^+ (6.5 nm) was determined from the indirect Fourier transformation of the experimental data (Figure 4B). The plot for Kbp- K^+ is characterized by a bell-shaped curve peaking at around $D_{max}/2$, indicative of a globular protein. In contrast, the apo-Kbp curve reveals some globularity with an extended tail region giving rise to higher r values, suggesting that the apoprotein is less globular in shape and may include an extended region. The D_{max} of apo-Kbp is 12.5 nm (Figure 4B). Foldedness and flexibility of Kbp in the presence and absence of K^+ was further analyzed by inspecting the normalized Kratky plot, which reports on a molecule's compactness and overall flexibility (Durand et al., 2010). In the normalized Kratky plot fully folded, globular proteins present a bell-shaped curve with a maximum close to 1.1. Kbp- K^+ possesses a maximum at approximately 1.36, indicative of globularity. In the case of apo-Kbp the peak maximum is shifted to the right (1.86) which, taken together with the increased area under the curve, indicates a non-globular, asymmetric character with substantial conformational heterogeneity within the protein (Figure 4C). SAXS data obtained for Kbp in the presence and absence of K^+ , were used to reconstruct low-resolution solution structures. Ab initio models were generated for each of apo-Kbp and Kbp- K^+ using GASBOR (Svergun et al., 2001) and within these, similar models were combined into an average model for each form of Kbp using DAMAVER (Volkov and Svergun, 2003) (Figure 4D).

To validate our structural models of both the high-resolution structure and *ab initio* generated models of Kbp-K⁺, we subjected the protein to hydrodynamic analysis using SOMO bead modeling and analytical ultracentrifugation experiments (Brookes et al., 2010) (Rai et al., 2005). Consistent with the SAXS data, which show a decrease in R_g upon K⁺ binding, sedimentation velocity experiments showed that K⁺ binding was associated with global structural changes in Kbp, with the Kbp-K⁺ complex ($s_{20,w}^0 = 1.09$ S) sedimenting considerably faster than apo-Kbp ($s_{20,w}^0 = 1.56$ S) due to a decrease in the frictional coefficient of the molecule (Figure S5 and Table S2). Experimentally obtained hydrodynamic parameters of Kbp were close to those derived from computed bead models of apo-Kbp and Kbp-K⁺ generated from SAXS *ab initio* models, validating these models (Table S2). Taken together, the SAXS and sedimentation velocity experiments suggest that Kbp undergoes a substantial compaction upon association with K⁺, possibly as a consequence of BON domain folding/ordering in the context of the entire molecule.

DISCUSSION

In this work we have shown that Kbp is an unprecedented, soluble K⁺ binding protein that is required by *E. coli* for normal growth at high levels of external K⁺. Kbp selectively binds a single K⁺ (radius = 1.33 Å) and larger alkali metal ions such as Rb⁺ (radius = 1.48 Å) and Cs⁺ (radius = 1.69 Å) but does not readily bind the smaller Na⁺ ion (radius = 0.95 Å), and in this respect has an ion specificity similar to that of K⁺ channels (Lockless et al., 2007). Kbp binds Rb⁺ with only slightly reduced affinity, but binding of the larger Cs⁺ ion is approximately an order of magnitude weaker. Binding of alkali metal ions is associated with a specific conformational change in Kbp, with NMR data indicating that in the absence of K⁺ Kbp is partially unfolded. This is consistent with DSC experiments that show a significant increase in the enthalpy of unfolding on K⁺ binding. Taken together with the phenotype of cells lacking Kbp, which display retarded growth at high potassium concentrations, the ability to selectively bind K⁺ suggests a role in potassium homeostasis. The phenotype of growth inhibition of cells lacking Kbp at high concentrations of K⁺ is remarkably similar to the phenotype displayed by gain-of-function mutants of the *E. coli* K⁺ channel, Kch (Kuo et al., 2003). These mutations are located in the cytoplasmic sensor domain of Kch and, when mapped onto the structure of this domain, form a putative ligand/protein binding site. Therefore, one possible explanation for the K⁺ sensitivity of the Δkbp strain is that Kbp is the regulatory protein of a potassium transport system.

Expression of the isolated domains showed that the BON domain retains the ability to bind K⁺, although with approximately 100-fold reduced affinity. Binding to the isolated LysM domain could not be detected. We therefore infer that, although the majority of groups involved in K⁺ binding are located in the BON domain, residues in the LysM domain must also be important in stabilizing the Kbp-K⁺ complex. Our data also suggest that the conformational change on K⁺ binding occurs predominantly in the BON domain, with NMR showing that in the absence of K⁺ the isolated BON domain is largely unfolded. The highly specific binding of K⁺ to an essentially unfolded region of poly-

peptide suggests that K⁺ specificity is determined in a way that is fundamentally different from that observed for K⁺ channels. In these membrane-bound channels, K⁺ binding sites are formed from backbone carbonyl groups from regions of polypeptide held in an unusual α -sheet conformation within the rigid constraints of the channel (Zhou et al., 2001). High-resolution structures of the K⁺ channel Ksc show K⁺ coordinated in the narrow selectivity filter by eight backbone carbonyl groups of the tetrameric transporter. It is thought that this topological arrangement is optimal for coordination of K⁺ and can therefore compensate for the energetic cost of desolvation on binding and transport (Bostick and Brooks, 2007). Selectivity for K⁺ over the smaller Na⁺ is imparted, as this topological arrangement does not effectively coordinate the smaller Na⁺ to the extent that the energetic penalty of desolvation cannot be adequately compensated. Outside the structural constraints of a channel there is little known about how soluble proteins are able to selectively bind K⁺ in preference to Na⁺. Our analysis of Kbp, even in conjunction with sequence comparison between Kbp homologs, does not yet reveal where K⁺ binds, and our work continues in this area.

Although we do not yet understand why Kbp is required for growth in media containing high levels of K⁺, one possibility is that it is acting as a direct sensor of cytoplasmic K⁺ concentration that subsequently enables adaptation to these conditions. The concentration of K⁺ in the cytoplasm of *E. coli* is typically 200 mM but can vary significantly, for example under osmotic stress. In addition, although the total concentration of K⁺ within the cytoplasm can be directly measured, there is considerable uncertainty as to how much of this exists in the free rather than the bound form, as the cytoplasm contains numerous species such as nucleic acids, phospholipids, protein, and small-molecule metabolites with an affinity for K⁺. In addition, ³⁹K NMR data suggest that up to 50% of cytoplasmic K⁺ may be in the bound form that is invisible to NMR (Richey et al., 1987). Clearly, if K⁺ were the only ligand of Kbp, a K_D of 160 μ M would preclude Kbp from this role, since at all feasible cytoplasmic K⁺ concentrations Kbp would be present only in the bound form. In addition, given the high level of selectivity for K⁺ it seems unlikely that an alternative ligand would directly compete with K⁺ for binding at the K⁺ binding site. However, if the LysM domain of Kbp is bound by a competing ligand and therefore cannot participate in stabilization of the BON domain-K⁺ complex, the affinity of Kbp for K⁺ may be reduced to that of the isolated BON-K⁺ complex, which has a K_D of approximately 20 mM. Given that the total concentration of K⁺ in the cytoplasm of *E. coli* is approximately 200 mM and there are high concentrations of competing ligands in the cell which will bind potassium, albeit with relatively low affinity, this is likely to be of the correct order of magnitude for a useful cytoplasmic K⁺ sensor.

One possible mechanism through which Kbp may regulate cation homeostasis is suggested by recent data indicating that the gene encoding Kbp and the downstream gene *yqaE*, which encodes a small membrane protein, are cotranscribed under conditions that induce the Cpx envelope stress response (Bernal-Cabas et al., 2015). Although the function of YqaE has not been elucidated, homologous proteins in yeast and plants, Pmp3 and RCI2 proteins, respectively, have been shown to play a key role in cation homeostasis and in particular the regulation of intracellular Na⁺ and K⁺ levels (Medina et al., 2001;

Rocha, 2016). Thus, given the genetic link between the genes encoding Kbp and YqaE, it is reasonable to speculate that in *E. coli* Kbp functions as a regulator of YqaE, which may act as a cation transporter and under stress conditions may play a role in cation homeostasis.

EXPERIMENTAL PROCEDURES

Cloning and Purification

Plasmid pKA1 was produced to express Kbp with a C-terminal His₆-tag (Kbp_{His6}). For construction of pCM6, *Kbp* from *E. coli* K-12 BW25113 was amplified using primers YgaUF (GGA ATT CCA TAT GGG TCT GTT CAA TTT TGT GAA AG) and YgaUR (CCG CTC GAG CTC TTC CGG AAT ACG CAA CAC TTG), and the PCR product was digested with NdeI and XhoI and introduced into the corresponding sites of pET21d (Novagen). Kbp_{His6} was overexpressed from *E. coli* BL21(DE3) carrying the plasmid pKA1. Five liters of Luria-Bertani (LB) broth was inoculated (1:100) and cultured overnight, and cells were grown at 37°C in a shaking incubator to an OD₆₀₀ = 0.6. Protein production was induced by the addition of 1 mM isopropyl β-D-1-thiogalactopyranoside, and the cells were grown for a further 3 hr and harvested by centrifugation. The cell pellet was resuspended in 20 mM Tris-HCl and 500 mM NaCl containing 5 mM imidazole (pH 7.5). Cells were lysed using an MSE Soniprep 150 (Wolf Laboratories) and the cell debris separated by centrifugation. The cell-free lysate was applied to a 5 ml HisTrap HP column (GE Healthcare) and the protein eluted over a 0- to 350-mM imidazole gradient. Kbp_{His6}-containing fractions were dialyzed overnight into 50 mM Tris-HCl and 50 mM NaCl (pH 7.5) and the protein was further purified by gel filtration chromatography on a Superdex S75 26/60 column (GE Healthcare) equilibrated in 50 mM Tris-HCl and 50 mM NaCl (pH 7.5). The protein was stored at -20°C in this buffer until required. For expression of the isolated Kbp BON domain (residues 1-89), plasmid pLJ1 was engineered by the introduction of an XhoI site in *kbp* from pKA1, digestion with XhoI to excise the XhoI fragment, and religation of the vector with T4 DNA ligase after gel purification. Site-directed mutagenesis of pAK1 to introduce an XhoI site was performed using the Stratagene QuikChange method with primers XhoIF (GTC GAT GAT CAG GTG AAA ACG CTC GAG CCA GCC ACT GCC AGC CAG) and XhoIR (CTG GCT GGC AGT GGC TGG CTC GAG CGT TTT CAC CTG ATC ATC GAC). For expression of the isolated Kbp LysM domain (residues 92-149), plasmid pLJ2 was engineered by the introduction of an NdeI site in *kbp* from pAK1, digestion with NdeI to excise the NdeI fragment, and religation of the vector with T4 DNA ligase after gel purification. Site-directed mutagenesis of pAK1 to introduce an NdeI site was performed using the Stratagene QuikChange method with primers NdeIF (GTC GAT GAT CAG GTG AAA ACG CAT ATG CCA GCC ACT GCC AGC CAG) and NdeIR (CTG GCT GGC AGT GGC TGG CAT ATG CGT TTT CAC CTG ATC ATC GAC). Kbp-Bon_{His6} and Kbp-LysM_{His6} were expressed in BL21(DE3) carrying plasmids pLJ1 and pLJ2, respectively, and purified as described for Kbp_{His6}. ¹⁵N-labeled proteins were expressed in BL21(DE3) grown in M9 minimal media with [¹⁵N]ammonium chloride as the sole nitrogen source.

NMR Spectroscopy

Kbp_{His6} was expressed in *E. coli* grown in M9 minimal medium with [¹⁵N] ammonium chloride and [¹³C₆]glucose as sole nitrogen and carbon sources. Samples were prepared containing approximately 1 mM Kbp_{His6} and 5 mM KCl in 20 mM NaPi (pH 7.2). Resonance assignments were made using a combination of standard double and triple resonance spectra and deposited (BMRB: 25839). Structures were calculated using ambiguously assigned distance restraints derived from crosspeaks picked in 3D ¹³C- and ¹⁵N-edited nuclear Overhauser effect spectroscopy-HSQC spectra (Marion et al., 1989; Vranken et al., 2005) recorded at 298 K. Phi and psi dihedral angle restraints for residues in regular secondary structure elements predicted by DANGLE (Cheung et al., 2010) were applied in the high-temperature dynamics phases of each structure calculation and omitted during the final cooling stage. In the final iteration of the ARIA 2.3-driven calculations (Habeck et al., 2004), a total of 5,626 restraints were used to generate 100 structures from which an ensemble of the 20 structures which agree most closely with the experimental restraints was selected and refined in a full force field including explicit solvent

(Linge et al., 2003). Structure quality parameters are summarized in Table 1. The structures and restraints are deposited in the protein data bank (PDB: 5FIM) and biological magnetic resonance data bank (BMRB: 25839).

Growth Assays

For determination of the effect of the addition of KCl into the growth media, cells were grown in a defined sodium phosphate media, NaM9, based on M9 minimal media with added KCl at the concentrations specified. The recipe for NaM9 salts is 3.4 g/l Na₂HPO₄·2H₂O, 6.8 g/l NaH₂PO₄, 0.5 g/l NaCl, and 1.0 g/l NH₄Cl. 50 ml of NaM9 salts was supplemented with MgSO₄, CaCl₂, and 2% glucose and inoculated with 0.5 ml of an overnight culture of wild-type or *Δkbp* BW25113 grown in LB broth. For experiments with added NaCl, cells were grown in standard M9 minimal media with additional NaCl as indicated.

Calorimetry

ITC experiments were performed on a VP-ITC microcalorimeter (MicroCal) in 50 mM Tris-HCl and 50 mM NaCl (pH 7.5) at 25°C with Kbp in the cell (cell volume = 1.4005 ml) and the MCl salts (where M is K⁺, Cs⁺, Rb⁺, NH₄⁺) in the syringe (10 mM). MCl was titrated into Kbp using an initial injection of 1 μl followed by 28 × 10-μl injections with stirring at 310 rpm. Prior to ITC measurements, proteins were dialyzed overnight against the dialysis buffer used for protein heat of dilution control experiments. Data for Kbp-M⁺ complex formation were fitted with a single-site binding equation after correction for heat of dilution of using MicroCal ORIGIN software. The thermal stability of apo-Kbp and the Kbp-K⁺ complexes was measured by DSC using a VP-DSC (MicroCal) at a scan rate of 60°C h⁻¹. Samples were degassed briefly prior to loading in the DSC cell. Data were analyzed using standard MicroCal ORIGIN instrumental software.

Circular Dichroism

Near-UV CD spectra were acquired on a Jasco J-810 spectropolarimeter in 50 mM Tris and 50 mM NaCl (pH 7.5) ± 10 mM KCl at a protein concentration of 100 μM. Far-UV CD spectra were acquired in 20 mM Tris and 20 mM Na₂HPO₄ (pH 7.5) ± 5 mM K₂HPO₄.

Stopped-Flow Fluorescence

Stopped-flow fluorescence experiments were performed on an Applied Photophysics stopped-flow spectrofluorimeter with a 1:1 mixing chamber. All experiments used an excitation wavelength of 280 nm with fluorescence monitored above 305 nm using a 305-nm cut-off filter. Entrance and exit slits were set to 0.7 mm. All experiments were performed in 50 mM sodium phosphate buffer (pH 7.5) at 24.1°C. For experiments using NaCl, protein was dialyzed into the sodium phosphate buffer with 0.3 M or 0.6 M NaCl overnight. A total of 5,000 data points were collected over the course of each reaction, with 12 runs averaged for each association measurement.

The association of Kbp with K⁺ was monitored under pseudo-first-order conditions. The concentration of Kbp (10 μM) was kept constant, varying the concentration of KCl dissolved in sodium phosphate buffer (0.5-1 mM) for the 0.05-M and 0.3-M NaCl measurements, and between 0.5 and 20 mM KCl for the 0.6-M NaCl measurements. All data were fitted with a single-exponential rate equation. Linear plots of k_{obs} were used to determine the bimolecular association rates for the complex formation.

Analytical Ultracentrifugation

Sedimentation velocity experiments were carried out in a Beckman Coulter Optima XL-I analytical ultracentrifuge using an An-50 Ti eight-hole rotor. Kbp at concentrations ranging from 12 μM to 60 μM in 50 mM Tris and 50 mM NaCl (pH 7.5) was loaded into 12-mm path-length charcoal-filled Epon double-sector centerpieces (360 μl per sample), and a 211-μM sample was loaded into a 3-mm path-length centerpiece (90 μl) and spun at 49,000 rpm for approximately 9 hr at 4°C. Scans were collected every 7 min using both interference and absorbance optics. The laser delay was adjusted prior to the run to obtain high-quality interference fringes. Absorbance data were acquired at 280 nm with a radial range of 5.8-7.2 cm and radial step size of 0.005 cm. Kbp in 5 mM KCl supplemented buffer was analyzed in exactly the same way. Data were analyzed using SEDFIT (Schuck, 2000) to obtain the apparent sedimentation coefficients of Kbp ± K⁺ using the continuous c(s) distribution model followed by fitting with the non-interacting discrete species model.

The concentration dependence of the apparent sedimentation coefficients were then extrapolated to infinite dilution to determine the concentration-independent $s_{20,w}^0$. SEDNTERP (Laue et al., 1992) was used to calculate the partial specific volume (from the amino acid sequence of Kbp), the buffer density, and viscosity at 20°C and 4°C.

Small-Angle X-Ray Scattering

SAXS was carried out on the X33 beamline at the Deutsches Elektronen Synchrotron. A range of concentrations (0.5–10 mg/ml) of Kbp with and without K⁺ were loaded into the mica sample holders and data were collected. Scattering curves for buffer alone were recorded before and after each protein sample and an average of the buffer scattering was subtracted from the sample scattering. In the case of the K⁺-bound form, the buffer was supplemented with 10 mM KCl. The logarithm of the scattering intensity, log I(q), was then plotted as a function of the scattering angle, q. The data obtained for each sample were analyzed using PRIMUS and GNOM (Konarev et al., 2003; Svergun, 1992). The distance distribution function, p(r), was obtained by indirect Fourier transform of the scattering intensity, computed in GNOM, to look for changes in D_{max} (the maximum separation between scattering partners within the molecule) and more subtle conformational changes. A Guinier plot (ln I(q) versus q²) was used to derive the radius of gyration, R_g, of Kbp, BON, and LysM ± K⁺ (Guinier, 1939).

ACCESSION NUMBERS

The structures and restraints are deposited in the protein data bank (PDB: 5FIM) and biological magnetic resonance data bank (BMRB: 25839).

SUPPLEMENTAL INFORMATION

Supplemental Information includes five figures and two tables and can be found with this article online at <http://dx.doi.org/10.1016/j.str.2016.03.017>.

AUTHOR CONTRIBUTIONS

K.U.A. and B.O.S. performed NMR analysis; I.J. and O.B. performed SAXS and AUC analysis; K.M., I.J., D.W., and S.M.K. performed other biophysical analyses; D.W. performed microbiological work; B.O.S. and D.W. conceived the work and wrote the manuscript with contributions from the other authors.

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